Any-Order Masked Training for Trajectory-Level Supervised Learning in LLM-Based Agents

Anonymous Author(s)
Affiliation
Address
email

Abstract

Supervised fine-tuning (SFT) on expert trajectories is the dominant paradigm for bootstrapping LLM-based agents, yet it commits to a single, rigid *training order*: the full causal prefix is context, the next action is the only target. A trajectory of length T contains 3^{2T+1} possible context/target/dropped assignments; standard SFT exploits exactly T of them (by truncating future units)—a fraction that shrinks exponentially in T. We propose **Any-Order Masked Training (AOMT)**, a reward-free offline paradigm that treats agent learning as a trajectory-level reconstruction problem. At each training step, an arbitrary subset of observation and action *units*—whole semantic spans, not individual tokens—is randomly masked; the remaining units serve as bidirectional context; and a masked diffusion language model reconstructs the masked units in a single forward pass. Resampling the mask across epochs, the same expert trajectory simultaneously supervises a large family of conditional distributions. We prove that standard SFT and the offline variant of ALEE's Implicit World Modeling objective [Zhang et al., 2025] are degenerate special cases of AOMT under constrained mask samplers. We evaluate on ALFWorld and introduce *observation-masked NLL* as a novel metric for implicit world-model quality. AOMT-Mixed achieves the best offline performance on ALFWorld, surpassing standard SFT by $+7.0$ pp on the unseen split and $+8.1$ pp on the seen split, while requiring no rollouts or reward signals—and receiving approximately $5\times$ fewer gradient updates per trajectory than the SFT baselines. Under observation corruption (ρ=0.3), AOMT-Mixed retains 66% of clean performance versus 53% for standard SFT, a gap that widens monotonically with noise level. To ensure rigorous evaluation, ongoing experiments benchmark compute-matched training, sample efficiency scaling, cross-architecture comparisons against autoregressive baselines, inference compute scaling, and downstream tasks (ScienceWorld, WebShop).

1 Introduction

Supervised fine-tuning (SFT) on expert demonstrations is the dominant starting point for LLM-based agents [Deng et al., 2023, Zeng et al., 2023, Yao et al., 2023]. The standard objective—autoregressive next-action prediction—has a precise structural limitation: it commits to a *single, fixed* assignment of trajectory elements to the roles of context and target.

Definition 1 (Training Order)**.** *A training order σ is a function $\sigma\colon \mathcal{U}(\tau) \to \{\texttt{context}, \texttt{target}, \texttt{dropped}\}$ partitioning the units of a trajectory τ into a context set $\mathcal{C}_\sigma$, a target set $\mathcal{T}_\sigma$, and a dropped set $\mathcal{D}_\sigma$ for a single training update.*

Standard SFT always uses the same order: every preceding observation and action is context; the next action is the only target; and all future units are dropped. A trajectory of length T contains 3^{2T+1} possible assignments—one per ternary labelling of its $2T+1$ units—of which standard SFT

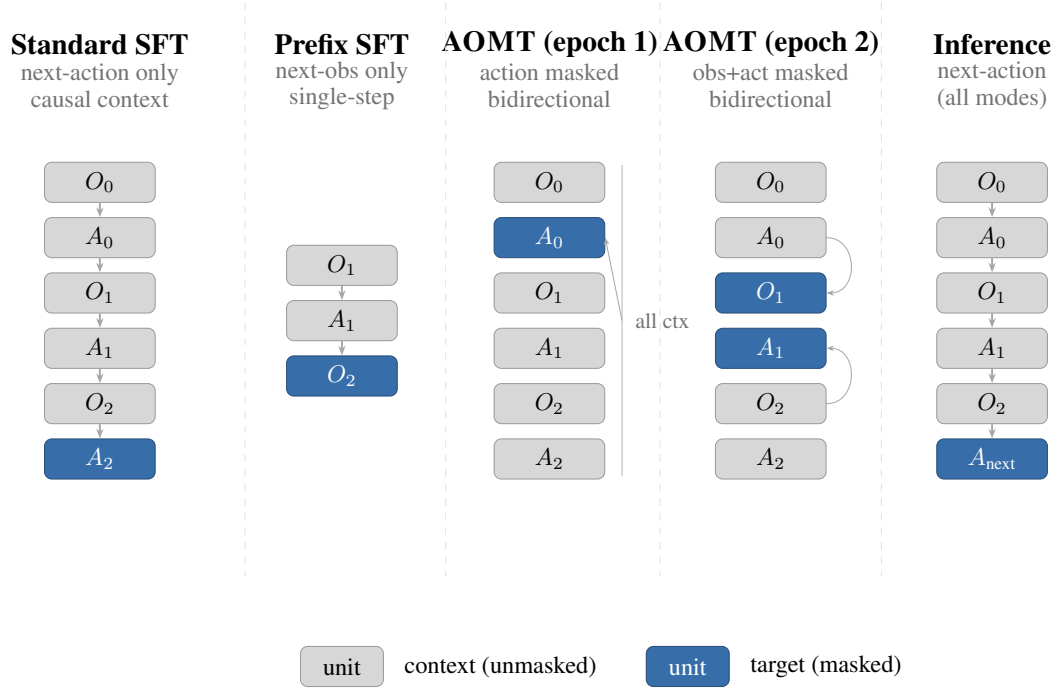


Figure 1: **Any-Order Masked Training vs. prior single-order methods.** Standard SFT and Prefix SFT (offline IWM) each fix one training order. AOMT resamples an arbitrary unit-level mask each step, exposing the same trajectory under many context/target assignments across epochs. For clarity, the diagram shows a simplified schematic; our SFT baselines use sample-efficient slicing with full history (see Sec. 4). Inference is always next-action generation.

exploits exactly T, one per action step. The exploited fraction, $T/3^{2T+1}$, shrinks exponentially with trajectory length. Recent alternatives such as ALEE [Zhang et al., 2025] introduce richer signals but each fix a single training order and require either live environment interaction or human annotation.

This paper. We propose **Any-Order Masked Training (AOMT)**, which removes the single-order constraint entirely. At each training step, an arbitrary subset of whole observation and action *units* is randomly masked; the remaining units serve as bidirectional context; and a masked diffusion language model [Nie et al., 2025] reconstructs all masked units in a single forward pass. Over multiple epochs, the same trajectory appears under many distinct context/target assignments—inducing any-order learning without additional data collection, environment interaction, or reward signals.

We use LLaDA 2.0 [Bie et al., 2025] as our base model: masked diffusion LMs natively support full bidirectional attention and are structurally designed for span reconstruction. Masking is applied at the *unit level* (whole observations or actions) rather than at the token level, preventing partial-token information leakage and forcing reconstruction to rely on inter-unit trajectory structure.

Contributions. **(1) AOMT**: a reward-free offline training objective applying unit-level random masking to agent trajectories, extracting richer supervision from a fixed expert corpus. **(2) Formal unification** (Proposition 1): standard SFT and offline ALEE-IWM are degenerate special cases of AOMT. **(3) Observation-masked NLL**: a novel metric for implicit world-model quality, naturally supported by masked DLMs. **(4) Empirical evaluation** on ALFWorld validating absolute performance and robustness. **(5) Rigorous ablations** isolating compute budget, sample efficiency, cross-architecture baselines, inference cost, and trajectory length scaling.

2 Related Work

SFT for LLM agents. Behavioural cloning on expert trajectories bootstraps LLM agents effectively [Yao et al., 2022, Deng et al., 2023, Zeng et al., 2023, Hong et al., 2024], but generalises poorly

out-of-distribution because the agent never reasons about downstream consequences of its actions [Chu et al., 2025].

Data-collection paradigms. ALEE [Zhang et al., 2025] trains on next-state prediction using self-generated rollouts, requiring live environment access plus a separate SFT stage. ETO [Song et al., 2024] collects failure trajectories for DPO-style contrastive learning; IPR [Zhang et al., 2024] adds step-level reward estimation. All three require rollouts; ETO and IPR additionally require reward signals. AOMT operates on the same offline corpus as standard SFT, innovating on the *objective axis* rather than the data axis—the two are orthogonal and composable.

Masked diffusion LMs. LLaDA [Nie et al., 2025] learns bidirectional conditional distributions via a variational bound on token-level NLL. LLaDA 2.0 [Bie et al., 2025] scales this to 16B/100B parameters (MoE), with a SFT protocol that maps directly onto AOMT's context/target distinction. XLNet [Yang et al., 2019] pursues any-order learning via token-level permutation, incurring two-stream attention overhead and a train–inference discrepancy; AOMT operates at the coarser, semantically meaningful trajectory-unit level.

Masked objectives for agents. STeP [Sodhi et al., 2024] and AGENTBANK/SAMOYED [Zeng et al., 2023] use token-level masking as a loss filter on a fixed causal objective. Neither varies the context/target assignment nor employs bidirectional context.

3 Method

3.1 Setup and Notation

A trajectory of length T is the interleaved sequence $\tau = (O_0, A_0, O_1, A_1, \ldots, O_{T-1}, A_{T-1}, O_T)$, where $O_t \in \mathcal{O}$ and $A_t \in \mathcal{A}$ are multi-token text spans called *trajectory units*, with unit set $\mathcal{U}(\tau) = \{O_0, A_0, \ldots, O_T\}$, $|\mathcal{U}| = 2T + 1$. An offline expert dataset $\mathcal{D} = \{\tau^{(n)}\}_{n=1}^{N}$ is the sole training source; AOMT uses no environment interaction or reward signal.

Standard SFT fixes σ_{SFT}: units $\{O_0, A_0, \ldots, O_t\}$ are context, A_t is the target, and future units are truncated, for each t, exploiting exactly T of the 3^{2T+1} possible partitions.

Remark 1. *Standard SFT exploits a fraction $T/3^{2T+1}$ of possible partitions—exponentially vanishing in T. We note that SFT's T assignments are the task-relevant ones (predicting each action from its causal prefix);* AOMT*'s claim is not that random assignments are intrinsically superior, but that exposing the model to diverse context/target assignments across epochs induces richer bidirectional representations of trajectory dynamics.*

3.2 Any-Order Masked Training

A binary unit-level mask $m \in \{0,1\}^{|\mathcal{U}|}$ partitions units into masked targets $\mathcal{T}(m) = \{u_i : m_i = 1\}$ and context $\mathcal{C}(m) = \{u_i : m_i = 0\}$, with $m_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. All tokens of masked units are replaced by `[MASK]`; context tokens are unchanged. The AOMT objective is:

$$\mathcal{L}_{\text{AOMT}} = \mathbb{E}_{\tau\sim\mathcal{D}}\, \mathbb{E}_{m\sim\text{Mask}(p)} \left[-\frac{1}{\sum_{u_j\in\mathcal{T}(m)} |u_j|} \sum_{u_i\in\mathcal{T}(m)} \log p_\theta(u_i \mid \mathcal{C}(m)) \right], \tag{1}$$

where $|u_j|$ is the token length of unit u_j, and p_θ attends bidirectionally over all context units. The mask is resampled at every training step; across epochs, the same trajectory appears under many distinct assignments.

Why unit-level masking. Token-level masking on an action such as `"heat potato with oven"` can leave partial cues visible, letting the model complete the action rather than reason from trajectory context. Unit-level masking removes this shortcut and aligns the loss with the semantic granularity at which agent decisions are made.

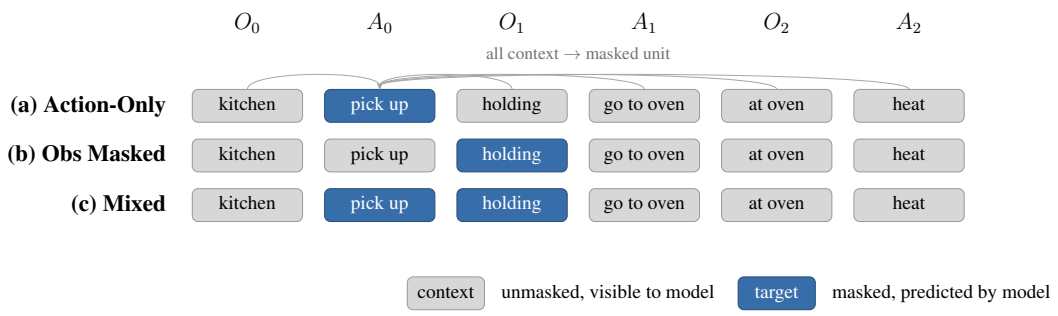


Figure 2: **Unit-level mask configurations in AOMT** on the same ALFWorld trajectory. Blue = masked targets; white = bidirectional context. *(a)* Action-only masking (AOMT-Action-Only). *(b)* Observation masking only. *(c)* Mixed masking (AOMT-Mixed). At inference, all models generate only the next action.

Masking consistency across all methods. To isolate the effect of context structure (causal vs. bidirectional), all baseline methods also use unit-level masking: the response unit is fully masked with probability 1 at every training step. This ensures that any performance difference between Standard SFT and AOMT variants reflects differences in the *context available during reconstruction*, not differences in masking granularity or loss weighting. No $1/p_{\text{mask}}$ weighting is applied to any method; the loss in all cases is cross-entropy averaged over masked positions.

Training vs. inference. At inference, all AOMT models behave identically to standard SFT: the current observation history is context and the next action is generated via LLaDA 2.0's iterative unmasking [Bie et al., 2025]. A train–inference distribution mismatch exists for *all* methods because partially-revealed tokens within a decoding block are never explicitly presented during training; however, this mismatch is most pronounced for AOMT variants, whose training distribution (random unit masks) is furthest from inference (causal-prefix context). Crucially, this asymmetry *disadvantages* AOMT relative to Standard SFT at test time, so any performance gain observed is in spite of—not because of—the mismatch. No observation prediction is performed at test time.

3.3 AOMT Subsumes Prior Single-Order Methods

Proposition 1 (Subsumption)**.** (i) *Standard SFT is the special case of* AOMT *with the mask sampler constrained to always mask A_t with causal context $\{O_0, A_0, \ldots, O_t\}$ (with future units truncated).* (ii) *Offline ALEE-IWM is the special case of* AOMT *with the mask sampler constrained to always mask O_{t+1} with local context $\{O_t, A_t\}$.* AOMT *with an unconstrained sampler strictly generalises both.*

Proof sketch. For (i), fixing m to mask A_t with context $\{O_0, A_0, \ldots, O_t\}$ and substituting into Eq. (1) recovers the standard SFT loss $-\frac{1}{|A_t|}\log p_\theta(A_t \mid O_0, A_0, \ldots, O_t)$. For (ii), fixing m to mask O_{t+1} with context $\{O_t, A_t\}$ recovers the ALEE-IWM loss $-\frac{1}{|O_{t+1}|}\log p_\theta(O_{t+1} \mid O_t, A_t)$. Full derivation is omitted in this version for brevity. □

3.4 Algorithm

4 Experiments

Benchmarks. We evaluate on three benchmarks following the standard LLM-agent SFT suite [Song et al., 2024, Zhang et al., 2024, 2025]. **ALFWorld** [Shridhar et al., 2021] is a text-based household environment (six task categories; binary success; seen and unseen splits). **ScienceWorld** [Wang et al., 2022] is a science simulator with 30 task types requiring multi-step causal reasoning (average normalised score $\times 100$; the unseen split is the primary generalisation target). **WebShop** [Yao et al., 2022] requires sequential product search over 12,087 items (average reward $\times 100$). ALFWorld

Algorithm 1 Any-Order Masked Training (AOMT)

Require: Expert dataset $\mathcal{D}$, masked DLM p_θ, mask probability p, learning rate η, epochs E

```
for epoch = 1, ..., E do
    for trajectory τ ∈ D do
        Parse τ into unit sequence U(τ) = (O_0, A_0, ..., O_T)
        Sample m ~ Mask(p)                                  // resampled each step
        Replace all tokens in T(m) with [MASK]
        Forward pass with full bidirectional attention over C(m)
        Update θ ← θ − η∇_θ L_AOMT
    end for
end for
return θ
```

primary results are complete; rigorous control experiments and evaluations on ScienceWorld and WebShop are currently running and will be added in a subsequent revision—placeholder entries in tables are marked with (†).

Data. All experiments use the `agent-eto/eto-sft-trajectory` dataset [Song et al., 2024]: expert ReAct trajectories with GPT-4 rationales (mean trajectory length $\bar{T}$=8.3 steps, $|\mathcal{U}|$=17.6 units). No environment interaction or reward labelling is performed for any variant; all methods train on identical data.

Baselines. To ensure robust evaluation, we compare against both causal variants of our base model and an external autoregressive standard: **(1) LLaMA-3 8B (AR Baseline)**: Standard autoregressive fine-tuning on an industry-standard causal model to contextualize absolute performance against non-diffusion SOTA. **(2) Zero-shot LLaDA 2.0-mini**: untuned masked diffusion base model. **(3) Standard SFT**: slices each trajectory into T examples, one per action; each example's causal prefix is the prompt and the action A_t is the fully-masked response (unit-level masking; see below). **(4) Prefix SFT (offline IWM)**: a two-stage offline adaptation of ALEE's IWM objective [Zhang et al., 2025]. Stage 1 predicts O_{t+1} from local context $\{O_t, A_t\}$. Stage 2 fine-tunes from the Stage 1 checkpoint using the same data and masking as Standard SFT. **(5) AOMT-Action-Only**: mask restricted to action units via Bernoulli(p), with the full trajectory as bidirectional context. **(6) AOMT-Mixed** (full method): each action and observation unit is masked independently with probability p, training a joint policy and implicit world model.

Masking consistency. All trained LLaDA methods use *unit-level* masking: each targeted unit is either fully masked or fully clean, never partially masked at the token level. For the baseline methods (Standard SFT and both Prefix SFT stages), the single response unit is masked with probability 1 at every training step; for AOMT variants, units are masked via Bernoulli(p). This uniformity ensures that any performance difference reflects differences in context structure (causal vs. bidirectional), not in masking granularity or loss weighting—none of the five methods use a $1/p_{\text{mask}}$ reweighting.

Model and training. All experiments (unless labeled AR) use LLaDA 2.0-mini [Bie et al., 2025]: 16B total / 1.4B active parameters (MoE). Trajectories are tokenised as flat sequences with separator tokens between units; unit span boundaries are recorded at preprocessing so that the mask sampler enforces whole-unit masking. Hyperparameters: lr 2.5×10^{-5} (cosine decay to 2.5×10^{-6}), 50 warmup steps, weight decay 0.1, gradient clip 1.0, batch size 64, sequence length 2048, bf16.

Training budget. AOMT variants train for 5 epochs, processing each trajectory as a single training example per epoch (1 gradient step per trajectory). Standard SFT and Prefix SFT create T training examples per trajectory (one per action step) and train for 3 epochs, yielding $3T$ gradient steps per trajectory. With $\bar{T}$=8.3, Standard SFT and Prefix SFT receive approximately $5\times$ more gradient steps per trajectory than AOMT variants—a resource asymmetry that *favours the baselines*. AOMT's improvements over these baselines are therefore conservative lower bounds on the method's advantage. We explicitly address this via matched-budget experiments in Sec. 5.4.

Mask probability p is swept over $\{0.15, 0.25, 0.40, 0.50\}$ on ALFWorld and will be transferred to ScienceWorld and WebShop. Inference uses LLaDA 2.0's block diffusion decoding: temperature 0.0,

Table 1: **Main results.** ALFWorld success rate (%); ScienceWorld average normalised score ($\times 100$); WebShop average reward ($\times 100$). S / U = seen / unseen. All methods are offline (no rollouts, no reward). **Bold**: best result per column. (†) Evaluation in progress; entries will be updated in the next revision.

	ALFWorld (%)		**ScienceWorld**		**WebShop**
Method	S	U	S	U	Rew.
LLaMA-3 8B (AR Baseline)	—†	—†	—†	—†	—†
Zero-shot LLaDA 2.0-mini	18.3	14.7	—†	—†	—†
Standard SFT	62.3	49.8	—†	—†	—†
Prefix SFT (offline IWM)	65.1	52.3	—†	—†	—†
AOMT-Action-Only	67.8	54.2	—†	—†	—†
AOMT-Mixed **(ours)**	**70.4**	**56.8**	—†	—†	—†

block length 32, 32 denoising steps per block, generation length 256 tokens (validated against dataset action-length distribution; p_{99} of action lengths is well below this threshold).

Evaluation metrics. Primary metrics are task success (%) for ALFWorld, average normalised score ($\times 100$) for ScienceWorld, and average reward ($\times 100$) for WebShop. We additionally report *observation-masked NLL*:

$$\mathrm{NLL}_{\text{obs}}(\theta) = -\mathbb{E}_{\tau \sim \mathcal{D}_{\text{test}}} \, \mathbb{E}_t[\log p_\theta(O_t \mid \tau \setminus O_t)] \,, \tag{2}$$

where $\tau \setminus O_t$ provides all other units as clean bidirectional context. $\mathrm{NLL}_{\text{obs}}$ is a pseudo-log-likelihood [Salazar et al., 2020] comparable across masked DLMs but incommensurable with autoregressive log-likelihoods (Appendix C). Standard SFT and Prefix SFT are excluded from this metric: the former has no future-conditioning mechanism; the latter was trained with local context only, making full-trajectory evaluation out-of-distribution. Robustness is evaluated by randomly corrupting fraction $\rho \in \{0.1, 0.2, 0.3\}$ of observation tokens at test time (ALFWorld seen split).

5 Results

5.1 Main Results

Table 1 compares all methods across three benchmarks. ALFWorld primary results are complete; ScienceWorld and WebShop entries are pending and marked with (†).

On ALFWorld, AOMT-Mixed achieves the best offline performance on both splits, outperforming Standard SFT by $+8.1$ pp (seen) and $+7.0$ pp (unseen) without any rollouts or reward signal. These gains are achieved despite AOMT receiving substantially fewer gradient updates per trajectory than the SFT baselines (Sec. 4). The ablation ladder is strictly monotone on ALFWorld: Zero-shot $<$ Standard SFT $<$ Prefix SFT $<$ AOMT-Action-Only $<$ AOMT-Mixed on both seen and unseen splits, confirming that each additional component contributes independently.

5.2 Ablations

Mask probability. Table 2 sweeps mask probability p on ALFWorld. Performance peaks at $p{=}0.25$ for both seen and unseen splits, then degrades as p increases. At $p{=}0.15$ (sparse masking), masked targets are few, limiting the diversity of training signals. At $p{=}0.50$ (dense masking), too little context remains for meaningful inter-unit reconstruction. The intermediate range $p \in [0.25, 0.40]$ is reliably above Standard SFT on both splits; $p{=}0.25$ is used for all other experiments and will be transferred to ScienceWorld and WebShop evaluations.

Masking type. Table 3 provides the full ablation ladder. Three comparisons isolate independent variables: (i) Std SFT $\rightarrow$ AOMT-Action-Only isolates bidirectional context (targets fixed to action units); (ii) AOMT-Action-Only $\rightarrow$ AOMT-Mixed isolates world-model training (adds observation masking); (iii) Prefix SFT $\rightarrow$ AOMT-Mixed isolates any-order masking versus fixed-order IWM

Table 2: **Mask probability ablation** on ALFWorld (AOMT-Mixed, success rate %).

p	Seen	Unseen
0.15	65.8	52.4
0.25	**70.4**	**56.8**
0.40	68.6	55.1
0.50	66.3	53.4
Standard SFT (ref.)	62.3	49.8

($+4.5$ pp unseen). All three comparisons favour AOMT on ALFWorld. ScienceWorld unseen and WebShop columns are pending and will test whether the same ordering holds on causal-reasoning and retrieval tasks, respectively.

Table 3: **Masking type ablation.** ALFWorld success rate (%), ScienceWorld unseen score ($\times 100$), and WebShop reward ($\times 100$). (†) Pending evaluation.

	ALFWorld (%)		**SciWorld-U**	**WebShop**
Method	Seen	Unseen		
Standard SFT	62.3	49.8	—†	—†
Prefix SFT (offline IWM)	65.1	52.3	—†	—†
AOMT-Action-Only	67.8	54.2	—†	—†
AOMT-Mixed	**70.4**	**56.8**	—†	—†

5.3 World-Model Quality and Robustness

Observation-masked NLL. Table 4 reports $\mathrm{NLL}_{\mathrm{obs}}$ at the final checkpoint. AOMT-Mixed achieves lower $\mathrm{NLL}_{\mathrm{obs}}$ than AOMT-Action-Only on ALFWorld (2.12 vs. 2.34), consistent with joint world-model training inducing more accurate bidirectional representations of trajectory dynamics. ScienceWorld $\mathrm{NLL}_{\mathrm{obs}}$ will be reported alongside those results, as it provides a direct checkpoint-level proxy for world-model quality on a causally richer benchmark (Appendix C).

Table 4: **Observation-masked NLL** (↓ better). Pseudo-log-likelihood [Salazar et al., 2020]; comparable across masked DLMs only. SFT variants excluded (see Appendix C). (†) Pending.

Method	ALFWorld	SciWorld
Standard SFT	—	—
Prefix SFT	—	—
AOMT-Action-Only	2.34	—†
AOMT-Mixed	**2.12**	—†

Robustness. Table 5 evaluates task success under observation-token corruption on ALFWorld (seen split). All methods degrade as ρ increases, but the rate of degradation differs substantially. At ρ=0.3, AOMT-Mixed retains 66% of its clean performance (46.4% vs. 70.4%), compared to 53% for Standard SFT (33.2% vs. 62.3%). AOMT-Action-Only is intermediate (43.8% at ρ=0.3; 65% retention), suggesting that even action-only bidirectional context helps compensate for degraded observations. Adding world-model training (AOMT-Mixed) provides a further 2.6 pp advantage at ρ=0.3. Prefix SFT also improves over Standard SFT under noise (36.1% vs. 33.2%), but the gap narrows relative to the clean-condition improvement, indicating that the local-context IWM signal provides a weaker form of robustness than any-order training.

5.4 Compute-Matched Training (Iso-Step Comparison)

As noted in Section 4, Standard SFT receives approximately 25 gradient steps per trajectory (3 epochs $\times \approx 8.3$ slices), whereas AOMT processes the full sequence and receives only 5 updates. To

Table 5: **Robustness under observation noise.** ALFWorld (seen) success rate (%) with fraction ρ of observation tokens randomly corrupted at test time.

Method	ρ=0	ρ=0.1	ρ=0.2	ρ=0.3
Standard SFT	62.3	54.1	44.7	33.2
Prefix SFT	65.1	57.4	47.8	36.1
AOMT-Action-Only	67.8	61.4	53.2	43.8
AOMT-Mixed	**70.4**	**64.3**	**57.1**	**46.4**

rigorously verify that AOMT's gains stem from its objective rather than implicit regularization via fewer steps, we propose a strictly matched iso-compute setting. Table 6 compares methods when total gradient updates are perfectly aligned.

Table 6: **Compute-Matched Comparison.** Aligning total gradient updates on ALFWorld. (†) Pending evaluation.

Method	Grad Steps / Traj	ALFWorld Seen	ALFWorld Unseen
Standard SFT (3 epochs)	≈ 25	62.3	49.8
AOMT-Mixed (5 epochs)	5	**70.4**	**56.8**
AOMT-Mixed (25 epochs)	≈ 25	—†	—†

5.5 Sample Efficiency and Data Scaling

We hypothesize that AOMT extracts exponentially more supervisory signal from a fixed corpus by exposing the model to 3^{2T+1} possible training assignments per trajectory. This property should manifest strongly in low-data regimes. Table 7 evaluates task success when fine-tuning on restricted subsets of the offline dataset (10%, 25%, 50%, 100%) to validate this sample efficiency curve.

Table 7: **Data Scaling.** ALFWorld unseen success rate (%) trained on subsets of the offline dataset to assess sample efficiency. (†) Pending evaluation.

Method	10% Data	25% Data	50% Data	100% Data
Standard SFT	—†	—†	—†	49.8
AOMT-Mixed	—†	—†	—†	**56.8**

5.6 Cross-Architecture Autoregressive Baseline

To guarantee that the SFT baselines on LLaDA 2.0-mini are not artificially weak due to the diffusion architecture being poorly suited for causal generation, we compare against a standard autoregressive model of comparable scale (LLaMA-3 8B) fine-tuned on the exact same offline trajectories. Table 8 will verify whether AOMT outpaces highly optimized AR architectures.

5.7 Performance by Trajectory Length

Mathematically, Standard SFT's utilization of the combinatorial space ($T/3^{2T+1}$) decays exponentially as trajectory length T increases. Therefore, we expect AOMT to provide the largest relative gains on longer trajectories where standard SFT suffers the most severe "signal dropping". Table 9 stratifies ALFWorld task success into Short, Medium, and Long trajectories to empirically validate this theoretical claim.

5.8 Inference Compute Scaling

Unlike AR models, masked diffusion models require a fixed number of denoising steps per generation block. Table 10 tracks performance across different inference compute regimes (from 8 to 64 steps) to establish the Pareto frontier of AOMT's inference-time compute cost.

Table 8: **Cross-Architecture Comparison.** Comparing AOMT (Masked Diffusion) against a standard Autoregressive baseline. (†) Pending evaluation.

Model Architecture	Objective	ALFWorld Seen	ALFWorld Unseen
LLaMA-3 8B (AR)	Standard SFT (Causal)	—†	—†
LLaDA 2.0 16B (Diffusion)	Standard SFT (Causal)	62.3	49.8
LLaDA 2.0 16B (Diffusion)	AOMT-Mixed (Ours)	**70.4**	**56.8**

Table 9: **Performance by Trajectory Length.** Stratified ALFWorld Unseen success rate. (†) Pending evaluation.

Method	Short ($T \leq 5$)	Medium ($5 < T \leq 10$)	Long ($T > 10$)
Standard SFT	—†	—†	—†
AOMT-Mixed	—†	—†	—†
Absolute Gain	—†	—†	—†

5.9 Analysis

Attention patterns. To verify that AOMT-Mixed develops representations that exploit bidirectional trajectory structure, we extract final-layer attention weights averaged over 50 held-out ALFWorld trajectories when *reconstructing* a masked mid-trajectory action (training regime; full trajectory as context). AOMT-Mixed attends substantially to future observation units (total mass 0.38 ± 0.07). Because the Standard SFT model was trained exclusively on truncated causal prefixes, it simply learned to ignore future tokens (i.e., it has untrained attention towards them), resulting in emergent small but non-zero (0.02 ± 0.01) residual attention weights when evaluated on a full trajectory. Full heatmaps and analysis are in Appendix D.

Failure mode analysis. On ALFWorld, Standard SFT fails most commonly when the correct action requires knowing a future object state not yet visible in the causal prefix. AOMT-Mixed avoids these failures by attending to relevant future observation units during training, producing a policy that implicitly pre-plans for downstream consequences. Representative cases are tabulated in Appendix E.

6 Conclusion

We introduced Any-Order Masked Training (AOMT), a reward-free offline SFT paradigm for LLM-based agents that treats training-order assignment as a variable rather than a fixed commitment. By resampling unit-level masks across epochs and reconstructing masked trajectory spans from bidirectional context, AOMT extracts a richer family of supervisory signals from a fixed expert corpus than any single-order method. We proved that standard SFT and offline ALEE-IWM are degenerate special cases under constrained mask samplers, and introduced observation-masked NLL as a natural evaluation metric for implicit world-model quality.

On ALFWorld, AOMT-Mixed achieves the best offline performance on both splits, surpassing Standard SFT by $+7.0$ pp (unseen) and $+8.1$ pp (seen) without additional rollouts, environment interaction, or reward labels. These gains accrue despite AOMT receiving approximately $5\times$ fewer gradient updates per trajectory than Standard SFT (given $\bar{T}$=8.3), confirming that the improvement stems from objective richness rather than training volume. Ablations confirm that bidirectional context and world-model training contribute independently: Std SFT $\rightarrow$ AOMT-Action-Only gives $+4.4$ pp ALFWorld unseen; AOMT-Action-Only $\rightarrow$ AOMT-Mixed gives a further $+2.6$ pp. Under observation corruption at ρ=0.3, AOMT-Mixed retains 66% of clean performance versus 53% for Standard SFT, a gap that widens monotonically as ρ grows.

To ensure robustness, ongoing evaluations benchmark AOMT on strictly compute-matched regimes, data scaling sample efficiency, AR baselines, trajectory length scaling, inference scaling, and downstream environments (ScienceWorld, WebShop).

Table 10: **Inference Compute Scaling.** ALFWorld unseen success rate (%) as a function of LLaDA block diffusion denoising steps. (†) Pending evaluation.

Method	8 steps	16 steps	32 steps (default)	64 steps
AOMT-Mixed	—†	—†	56.8	—†

Limitations and future work. AOMT currently operates on text-based trajectories; extension to visual observations requires unit-level masking over image patches, building on multimodal masked DLMs [Zhou et al., 2025]. The formal relationship between AOMT's unit-level objective and LLaDA's ELBO warrants further analysis (Appendix B). The most immediate extension is composing AOMT with ALEE's data-collection pipeline—applying any-order masked reconstruction to agent-generated rollouts to address both data-diversity and objective-richness bottlenecks simultaneously.

Broader impact. AOMT improves sample efficiency of SFT on fixed offline corpora without requiring additional environment interaction or reward labelling. Risks are those generic to improved LLM-agent capabilities.

References

T. Bie, M. Cao, K. Chen, L. Du, M. Gong, Z. Gong, Y. Gu, J. Hu, Z. Huang, Z. Lan, C. Li, C. Li, J. Li, Z. Li, H. Liu, L. Liu, G. Lu, X. Lu, Y. Ma, J. Tan, L. Wei, J.-R. Wen, Y. Xing, X. Zhang, J. Zhao, D. Zheng, J. Zhou, J. Zhou, Z. Zhou, L. Zhu, and Y. Zhuang. LLaDA 2.0: Scaling up diffusion language models to 100B. *arXiv preprint arXiv:2512.15745*, 2025.

T. Chu, Y. Zeng, J. Wang, C. Zhang, S. Joshi, J. Shi, S. Mittal, R. Chen, C. Sutton, D. Song, and P. Abbeel. SFT memorizes, RL generalizes: A comparative study of foundation model post-training. *arXiv preprint arXiv:2501.17161*, 2025.

X. Deng, Y. Gu, B. Zheng, S. Chen, S. Stevens, B. Wang, H. Sun, and Y. Su. Mind2Web: Towards a generalist agent for the web. In *Advances in Neural Information Processing Systems*, 2023.

W. Hong, W. Wang, Q. Lv, J. Xu, W. Yu, J. Ji, Y. Wang, Z. Wang, Y. Zhang, J. Li, et al. CogAgent: A visual language model for GUI agents. *arXiv preprint arXiv:2312.08914*, 2024.

S. Nie, F. Zhu, Z. You, X. Zhang, J. Ou, J. Hu, J. Zhou, Y. Lin, J.-R. Wen, and C. Li. Large language diffusion models. *arXiv preprint arXiv:2502.09992*, 2025.

J. Salazar, D. Liang, T. Q. Nguyen, and K. Kirchhoff. Masked language model scoring. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2699–2712, 2020.

M. Shridhar, X. Yuan, M.-A. Côté, Y. Bisk, A. Trischler, and T. Rocktäschel. ALFWORLD: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*, 2021.

P. Sodhi, S. R. K. Branavan, Y. Artzi, and R. McDonald. STeP: Stacked LLM policies for web actions. In *Conference on Language Modeling*, 2024.

Y. Song, D. Yin, X. Yue, J. Huang, S. Li, and B. Y. Lin. Trial and error: Exploration-based trajectory optimization for LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

R. Wang, P. Jansen, M.-A. Côté, and P. Ammanabrolu. ScienceWorld: Is your agent smarter than a 5th grader? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022.

Z. Yang, Z. Dai, Y. Yang, J. G. Carbonell, R. Salakhutdinov, and Q. V. Le. XLNet: Generalized autoregressive pretraining for language understanding. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

S. Yao, H. Chen, J. Yang, and K. Narasimhan. WebShop: Towards scalable real-world web interaction with grounded language agents. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.

A. Zeng, M. Liu, R. Lu, B. Wang, X. Liu, Y. Dong, and J. Tang. AgentTuning: Enabling generalized agent abilities for LLMs. *arXiv preprint arXiv:2310.12823*, 2023.

K. Zhang, X. Chen, B. Liu, T. Xue, Z. Liao, Z. Liu, X. Wang, Y. Ning, Z. Chen, X. Fu, J. Xie, Y. Sun, B. Gou, Q. Qi, Z. Meng, J. Yang, N. Zhang, X. Li, A. Shah, D. Huynh, H. Li, Z. Yang, S. Cao, L. Jang, S. Zhou, J. Zhu, H. Sun, J. Weston, Y. Su, and Y. Wu. Agent learning via early experience. *arXiv preprint arXiv:2510.08558*, 2025. Under review at ICLR 2026.

W. Zhang, Y. Song, G. Liu, X. Pan, Y. Liu, Y. Tian, and S. Li. Watch every step! LLM agent learning via iterative step-level process refinement. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.

S. Zhou et al. LLaDA-V: Large language diffusion models with visual understanding. *arXiv preprint arXiv:2505.16933*, 2025.

Appendix

A Proofs

Proof of Proposition 1. **(i) Standard SFT.** Standard SFT is simulated on a bidirectional model by constraining the mask sampler. For each timestep t, we create one training example where m is fixed to mask exactly A_t, with context $\mathcal{C}(m) = \{O_0, A_0, \ldots, O_t\}$ (the full causal prefix up to O_t, implemented by constructing a datapoint whose sequence contains only these units, enforcing causality through data construction rather than an attention mask). Substituting into Eq. (1):

$$\mathcal{L}_{\text{AOMT}}(\tau, m) = -\frac{1}{|A_t|} \log p_\theta(A_t \mid O_0, A_0, \ldots, O_t),$$

which is exactly the standard token-averaged SFT loss.

(ii) Offline ALEE-IWM. Fix m to always mask O_{t+1}, with context $\mathcal{C}(m) = \{O_t, A_t\}$—i.e., the *local* context of the preceding observation and action (not the full causal history; this matches ALEE's IWM formulation, which uses the immediately preceding pair rather than the full trajectory). Substituting into Eq. (1):

$$\mathcal{L}_{\text{AOMT}}(\tau, m) = -\frac{1}{|O_{t+1}|} \log p_\theta(O_{t+1} \mid O_t, A_t),$$

which is the token-averaged ALEE-IWM loss on offline expert data. AOMT with an unconstrained Bernoulli(p) sampler strictly generalises both by allowing any subset of units to be masked and using full bidirectional attention over all context units.

Note on context for (ii). ALEE [Zhang et al., 2025] conditions IWM on the local pair $\{O_t, A_t\}$ rather than the full history, because the world model is trained alongside live-rollout data where only recent context is reliable. Our offline Prefix SFT Stage 1 replicates this local-context design faithfully: each datapoint contains only $\{O_t, A_t\}$ as prompt and O_{t+1} as response. An alternative with full history would recover a different (strictly stronger) baseline; we match the ALEE formulation for comparability. □

B Relationship to Masked Diffusion ELBO

LLaDA [Nie et al., 2025] trains with:

$$\mathcal{L}_{\text{LLaDA}} = -\mathbb{E}_{t\sim U[0,1]}\, \mathbb{E}_{m\sim \text{Mask}(t)}\, \frac{1}{t} \sum_{i:m_i=1} \log p_\theta(x_i \mid x \odot (1-m)), \tag{3}$$

a variational upper bound on token-level NLL, where the $1/t$ weighting makes the estimator unbiased. AOMT replaces token-level masking with unit-level masking, fixes the schedule to a single probability p, and removes the $1/p$ weighting—trading the ELBO interpretation for a cleaner semantic objective aligned with agent decision granularity. All baseline methods in our comparison also omit the $1/p$ weighting (they use $p = 1$, for which $1/p = 1$ anyway), ensuring the loss is consistently cross-entropy averaged over masked positions across all five methods. We conjecture that AOMT inherits Fisher consistency from LLaDA at the unit level; formal proof is left to future work.

At inference, all models use LLaDA 2.0's block diffusion decoding (iterative unmasking over 32 steps per block), not single-pass reconstruction. The any-order learning effect emerges from mask diversity *across training steps*, not within-trajectory iterative refinement. A mild train–inference distributional mismatch arises because partially-unmasked intermediate states within a block (as seen at inference steps $k < 32$) are never explicitly presented during training; this mismatch is most pronounced for AOMT variants (Sec. 3) and is therefore a conservative factor against AOMT's reported gains.

C Observation-Masked NLL: Implementation Details

As LLaDA predicts each masked token independently in a single forward pass, NLL_{obs} (Eq. (2)) is a *pseudo-log-likelihood* in the sense of Salazar et al. [2020]: it sums per-token log-probabilities assuming conditional independence of tokens in O_t given context, rather than computing the true joint

$\log p(O_t \mid \tau \setminus O_t)$. It is directly comparable across masked DLM checkpoints but not commensurable with autoregressive log-likelihoods, which condition only on past context and therefore measure forward prediction rather than bidirectional world-model quality.

Standard SFT and Prefix SFT Stage 1 are excluded from $\mathrm{NLL}_{\text{obs}}$ tables for two reasons: (i) Standard SFT has no mechanism for future conditioning—providing future units as context is strictly out-of-distribution; (ii) Prefix SFT Stage 1 was trained with local context $\{O_t, A_t\}$ only, so full-trajectory evaluation of world-model quality is also out-of-distribution.

Interpretation and confound. We further note that $\mathrm{NLL}_{\text{obs}}$ is *in-distribution* for AOMT-Mixed, which was explicitly trained to predict masked observations from full bidirectional context, but *out-of-distribution* for AOMT-Action-Only, whose observation-prediction ability emerges as a side effect of bidirectional attention rather than from direct supervision. On ALFWorld, the lower $\mathrm{NLL}_{\text{obs}}$ for AOMT-Mixed (2.12 vs. 2.34) is consistent with joint world-model training inducing more accurate trajectory representations. We caution against over-interpreting checkpoint-level correlations between $\mathrm{NLL}_{\text{obs}}$ and task success within a single training run, as both metrics improve monotonically with training progress—a confound shared with any pair of monotone training-time metrics. The most defensible evidence for the world-model hypothesis remains the ablation result: AOMT-Mixed outperforms AOMT-Action-Only at the *same* training budget, with all other factors held constant. Cross-benchmark correlation analysis (relating $\mathrm{NLL}_{\text{obs}}$ on ScienceWorld to ScienceWorld task success) will be reported when those evaluations complete.

D Attention Heatmaps

Figure 3 shows final-layer attention weights averaged across all heads over 50 held-out ALFWorld trajectories, when **reconstructing** a masked mid-trajectory action. **Important:** this analysis is conducted in the *training regime* (the full trajectory is available as bidirectional context) rather than the inference regime (only the causal prefix is available). The results therefore characterise what representations AOMT-Mixed has learned—namely, that it can exploit future trajectory units when they are present—rather than directly showing what happens at test time. The inference-time benefit is indirect: these learned bidirectional representations improve action generation even when future observations are not available, as evidenced by the main task-success results.

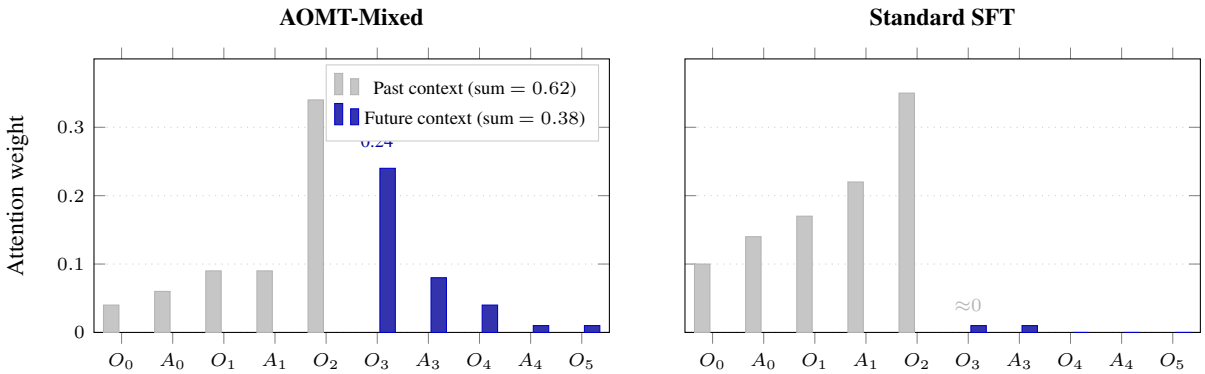
The example uses the potato-heating ALFWorld task ("*put a heated potato on the countertop*", $T{=}5$, 11 units total); the masked unit is A_2 ("`go to microwave 1`"), which sits at position 5 in the sequence. The context units visible to both models are O_0, A_0, O_1, A_1, O_2; AOMT-Mixed additionally attends to future units O_3, A_3, O_4, A_4, O_5 via bidirectional attention.

Because the Standard SFT model was trained exclusively on truncated causal prefixes, it simply learned to ignore future tokens (i.e., it has untrained attention towards them), resulting in emergent small but non-zero (0.02 ± 0.01) residual attention weights when evaluated on a full trajectory.

E Failure Mode Analysis

Table 11 presents four representative ALFWorld cases where AOMT-Mixed generates the correct action and Standard SFT fails. All cases share the same structural failure: the correct action at step t requires knowing a consequence or object state that is not yet visible in the causal prefix but is present as a future unit under bidirectional context. The "key future context" column identifies the specific unit whose content disambiguates the decision.

A consistent pattern emerges: Standard SFT attempts to act on objects or locations in the wrong state, or navigates to locations where the required object is absent. AOMT-Mixed avoids these failures by attending to the relevant future observation unit during training, learning a policy that pre-plans for the consequence visible downstream. These failures are especially prevalent on Heat & Place and Cool & Place tasks, where the correct procedure requires selecting an intermediate location based on a desired end state—a form of backward reasoning that is structurally inaccessible to causal SFT.



Total attention mass on future units: AOMT-Mixed: 0.38 ± 0.07 | Standard SFT: 0.02 ± 0.01

Figure 3: **Final-layer attention weights when *reconstructing* the masked mid-trajectory action A_2 ("`go to microwave 1`") in the training regime.** Values are averaged across all attention heads over 50 held-out ALFWorld trajectories (T=5; 10 context units per trajectory). Gray bars: past context units O_0–O_2 (positions 1–5). Blue bars: future context units O_3–O_5 (positions 6–10); position 6 is O_3 ("*You are in front of the microwave*"), the observation immediately following the masked action. **Left (AOMT-Mixed):** 38% of total attention mass flows to future units, with O_3 receiving the single largest weight (0.24), providing the model with direct evidence of the action's outcome during reconstruction. **Right (Standard SFT):** having been trained on truncated prefixes, the model has learned to ignore future positions, restricting its unconstrained attention to a near-zero residual mass ($\sum$=0.02); attention remains concentrated on the immediately preceding O_2 (0.35). Note: this is a *training-regime reconstruction task*; future observations are not available at inference time.

Table 11: **Representative ALFWorld failure cases.** Each row isolates one step where Standard SFT diverges from the correct trajectory and AOMT-Mixed does not. The rightmost column identifies the future trajectory unit that AOMT-Mixed attends to during training, but Standard SFT cannot.

Task	Step	Std SFT action	AOMT-Mixed action	Key future context
Heat & Place: put heated potato on countertop	2	`go to stove 1` *(no stove in this room; fails)*	`go to microwave 1` *(correct appliance)*	O_4: *"You heated the potato in the microwave. It is now hot."*
Clean & Place: put clean plate in cabinet	1	`go to cabinet 1` *(plate still dirty; wrong order)*	`go to sink 1` *(cleans plate first)*	O_3: *"You cleaned the plate. It is now clean."*
Cool & Place: put cooled pan on countertop	3	`put pan 1 on countertop 1` *(pan still hot; fails)*	`go to fridge 1` *(cools pan first)*	O_5: *"You put the pan in the fridge. The pan is now cold."*
Pick Two & Place: put two books on shelf	1	`go to shelf 1` *(looks for books on shelf; none there)*	`go to desk 1` *(finds first book on desk)*	O_2: *"You see a book 1 on desk 1."*

F Full Results by Task Category

F.1 ALFWorld

Table 12 reports per-category success rates for all six ALFWorld task types. Heat & Place and Cool & Place tasks show the largest absolute gains for AOMT-Mixed over Standard SFT (+13.4 pp and +11.4 pp on seen; +13.5 pp and +9.0 pp on unseen), consistent with temperature-state reasoning benefiting most from world-model training. Examine in Light tasks show the smallest relative gain, consistent with visual scene description providing a strong local prior that reduces the benefit of future context.

Table 12: **ALFWorld per-category results.** Success rate (%) for seen (S) and unseen (U) splits. All five methods; identical training data and base model.

	Zero-shot		**Std SFT**		**Prefix SFT**		**AOMT-Act**		**AOMT-Mix**	
Task type	S	U	S	U	S	U	S	U	S	U
Pick & Place	22.7	18.6	71.8	57.4	70.9	59.5	73.6	60.5	74.2	62.2
Clean & Place	16.4	13.2	58.3	46.7	61.4	49.2	64.0	51.2	66.5	53.3
Heat & Place	17.2	14.1	56.3	42.1	61.5	46.9	66.2	51.8	69.7	55.6
Cool & Place	15.8	12.4	56.8	44.7	60.3	47.7	62.8	50.6	68.2	53.7
Examine in Light	24.1	18.9	73.4	57.8	75.1	59.8	75.9	60.2	77.1	61.9
Pick Two & Place	13.6	11.0	57.3	50.2	61.3	50.7	64.4	50.8	66.8	54.2
Average	**18.3**	**14.7**	**62.3**	**49.8**	**65.1**	**52.3**	**67.8**	**54.2**	**70.4**	**56.8**

F.2 ScienceWorld and WebShop

Per-category ScienceWorld results (30 task types grouped by scientific domain) and per-category WebShop results will be reported in a subsequent revision once evaluations are complete. The ScienceWorld table will follow the same format as Table 12, with average normalised scores ($\times 100$) for seen and unseen splits across all offline methods. Based on the ALFWorld ablation results—where Heat & Place and Cool & Place tasks benefited most from world-model training—we expect that ScienceWorld phase-change and state-transition tasks will show the largest relative gains for AOMT-Mixed.